Synthetic Data Augmentation for Breast Cancer Diagnosis Using a Conditional Variational Autoencoder and Bayesian Logistic Regression

Tomohiro Oga, Gavin Kornitsky | Github
DS4420

Abstract

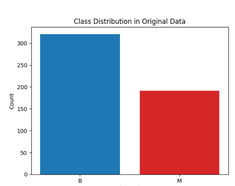
- Medical datasets are often **limited** because of privacy, consent, and regulations
- We investigate whether **synthetic data generation** can appropriately address this problem in breast cancer diagnosis
- We train a **Conditional Variational Autoencoder (CVAE)** to generate synthetic tabular samples and evaluate them with a Bayesian logistic regression classifier.
- Our findings show that we can use synthetically generated data to **improve an already strong predictive baseline**

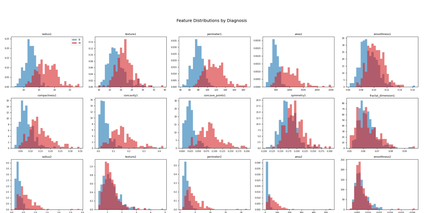
Research Question

This project asks whether a class-conditional VAE can generate synthetic data that can be utilized in downstream breast cancer classification to improve performance on held-out real data. In particular, we explore whether deep-learning based synthetic augmentation can improve performance beyond a strong baseline, whether controlled generation of class-conditioned samples can correct for the malignancy-scarce class-imbalance, and whether the learned latent space preserves diagnostically meaningful structure.

Dataset

- We used the **Wisconsin Diagnostic Breast Cancer (WDBC)** dataset, which contains 569 observations and 30 continuous features derived from fine-needle aspirates
- The task is a binary classification task between **benign** and **malignant** cases.
- This dataset is a clinically well-studied, tabular dataset, and is sufficiently small where augmentation is largely relevant.

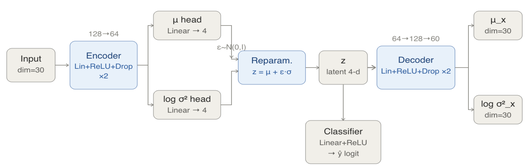




Motivation

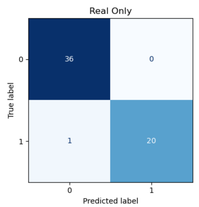
A standard VAE learns a model of p(x), where if we sample from it, we obtain more observations from the overall learned distribution, but there is no guarantee over whether generated samples are benign of malignant. For augmentation, this matters as our training data had more benign than malignant cases, but the clinically important errors are **false negatives** on malignant cases. This means that useful augmentation requires controlled/conditional generation from p(x | y) rather than only p(x). To this end, we used a class-conditional VAE with a parameterized conditional latent prior, a supervised latent label discriminator, gaussian negative log-likelihood reconstruction for continuous features, and KL scheduling during training.

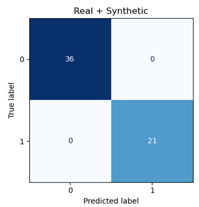
Methodology



- Our baseline model was a **Bayesian logistic regression**, implemented manually and trained on standardized WDBC features. It was evaluated on accuracy, precision, recall, F1, and AU-ROC
- Our generative model was a **class-conditional variational autoencoder.** The encoder mapped each 30-dimensional feature vector to an approximate posterior over a latent variable *z*, and the decoder predicted both reconstructed feature means and feature variances → the model both predicts both expected value and expected error given a latent *z*
- We include a **discriminator** conditioned on the latent space so that it preserves structure of diagnosis, then just being reconstructive.
- A KL warmup/annealing scheduler was introduced to stabilize training and to avoid premature posterior collapse
- The final **training objective** balances accurate feature reconstruction, diagnosis-aware latent structure, and regularization toward a parameterized class-specific prior
- The original dataset was **class imbalanced** (more benign points than malignant). We used the decoder from the conditional VAE to generate synthetic data in a controlled way by sampling using the learned priors
- The purpose of generation was not just to expand the training set but also to rebalance it → yielding ~50/50 class balance while effectively doubling the dataset

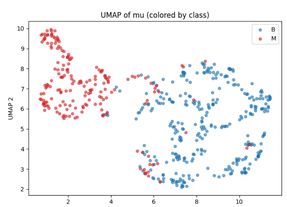
Results

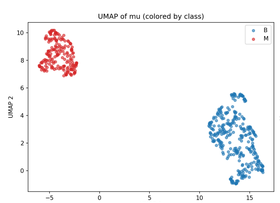




	Baseline	Augmented
Accuracy	0.982	**1.000**
Precision	1.000	**1.000**
Recall	0.952	**1.000**
F1 Score	0.976	**1.000**
AU-ROC	0.993	**1.000**

- The predictive performance of the baseline model is already **incredibly strong** with almost perfect performance except for one false negative
- Synthetic data insertion fixes this issue, where at least with this dataset, we then consequentially get perfect performance
- Additionally, the introduction of the class-based discriminative loss helped structure the latent representation such that we get tight class-based clustering





Conclusions and Next Steps

- Synthetic data generation was designed not only to increase the size of the training set but also to rebalance it, which may have contributed to the observed improvement in recall
- We see that without the class-based discriminative loss, the embedding space is only structured enough to maintain the most salient features for reconstruction, that (while co-related) may not promote class-based structure.
- In the future, we could do the same exact experimental setup with a larger dataset without as clean feature separation.
- Additionally, logistic regression is a low-capacity linear model, and may not benefit as much from synthetic augmentation, as much as a higher-capacity model like a neural network that is more prone to overfitting in low-data settings, and may benefit more from high-quality synthetic data.